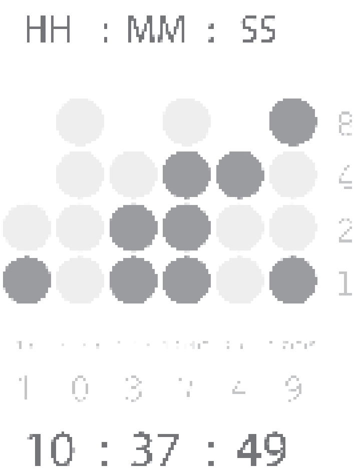
of the first battery in 1798 by Alessandro Volta, batteries have become a common power source for many household and industrial applications – so much so that the worldwide battery industry generates approximately $48 billion in sales each year.

Batteries work due to a chemical reaction inside them that causes electrons to flow from the negative to the positive terminal of the battery. When all of the chemicals inside the battery have reacted, the battery is spent. This is true for disposable batteries still in their original packaging as well with each losing 8-20% of their original charge every year simply due to being stored in warm temperatures of 20-30 degrees celcius. The same goes for rechargeable batteries, which lose a percentage of their charge each day when left off the charger and, despite the capability to be charged repeatedly, will eventually die.

The battery is one of mankind's top inventions because it is due to the advances in battery technologies that people no longer have to carry large brick phones and can now operate their smartphones for over 12-14 hours on a single charge.

Binary System



While most of the modern inventions are credited to the western world, the earliest known use of binary system is in the works of the Indian scholar Pingala. Although it's easiest for us humans to count in multiples of tens, electronic machines find using the binary system easier. It's easy because the electric/electronic medium uses the flow of electricity to indicate the 'on/off' state. Assuming the flow of electricity to be 1 and the lack of same to be 0, the electronic machines could be taught how to count using these two digits.

The roots of the binary system can be traced way back to the 'boolean' algebra – created by George Boole (whose